

MULTIMODAL K=6

For a fixed sample i and mode k , the predicted trajectory is a 12×2 vector

$$\textcircled{1} \hat{y}_{i,k} \in \mathbb{R}^{12 \times 2} = \hat{Y}_{i,k} = \begin{bmatrix} \hat{x}_{i,k,1} & \hat{y}_{i,k,1} \\ \hat{x}_{i,k,2} & \hat{y}_{i,k,2} \\ \hat{x}_{i,k,3} & \hat{y}_{i,k,3} \\ \vdots & \vdots \\ \hat{x}_{i,k,12} & \hat{y}_{i,k,12} \end{bmatrix} \quad \text{and GT} = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_{12} & y_{12} \end{bmatrix}$$

$Y \in \mathbb{R}^{12 \times 2}$

$\textcircled{2}$ then for every timestep t , collapse the (x,y) into Euclidean distance: $d_{i,k,t}$

$$d_{i,k,t} = \sqrt{(\hat{x}_{i,k,t} - x_{i,t})^2 + (\hat{y}_{i,k,t} - y_{i,t})^2} \Rightarrow d_{i,k} = \begin{bmatrix} d_{i,k,1} \\ d_{i,k,2} \\ d_{i,k,3} \\ \vdots \\ d_{i,k,12} \end{bmatrix}$$

for each mode k , sample i in B and future timestep t

for one sample and one mode

$\textcircled{3}$ then for this sample i , and mode k , use $d_{i,k}$ and SUM and divide by 12 to get the $ADE_{i,k} = \frac{1}{12} \sum_{t=1}^{12} d_{i,k,t}$

take average of

$$d_{i,k} = (d_{i,k,1} + d_{i,k,2} + d_{i,k,3} + \dots + d_{i,k,12}) / 12 \Rightarrow ADE_{i,k} \begin{cases} \text{for sample } i \\ \text{for mode } k \end{cases}$$

$\textcircled{3.1}$ NOW for this fixed sample i , repeat $\textcircled{1} \textcircled{2} \textcircled{3}$ for every MODE k

to get $ADE_{i,1} = 5.2$ e.g (mode 1)

$ADE_{i,2} = 3.1$ e.g (mode 2)

$ADE_{i,3} = 1.4$ e.g (mode 3)

$\vdots$

$ADE_{i,6} = 2.5$ e.g (mode 6)

$k \in \{1, 2, 3, 4, 5, 6\}$

$\in \{1..6\}$
↑

for this sample i in batch B

$\textcircled{4}$ select the min $\left[d_i = \min_{k \in \{1..6\}} ADE_{i,k} \right] \Rightarrow \text{sample } i \text{ 's } ADE_{i,3} = d_i$

sample y $ADE_{y,2} = d_y$

$\textcircled{5}$ compute d_i for every sample i in batch B .

⑤ so for one batch B with $i \in \{1 \dots 64\}$ samples we get

$$\underbrace{J_1 = \min_{k \in \{1 \dots 64\}} ADE_1}_{\sim} \text{ and } \underbrace{d_2 = \min_{k \in \{1 \dots 64\}} ADE_2}_{\sim} \dots \text{ until } \underbrace{J_{64} = \min_{k \in \{1 \dots 64\}} ADE_{64}}_{\sim}$$

then sum/take the average $\rightarrow \left[L_B = \frac{1}{B} \sum_{i=1}^{B=64} L_i \right]$

as: $\left[L_B = \frac{J_1 + J_2 + J_3 + \dots + J_{64}}{64} \right]$

Backprop
loss

the whole dimension of sequence:

$$[64, 6, 12, 2] \rightarrow [64, 6, 12] \rightarrow [64, 6] \rightarrow [64] \rightarrow [1]$$